GROUP WORK CAT 2 · DATABASE DESIGN

Student Management System

Database design · XAMPP and MySQL

Project scope

This database stores student details, courses, attendance, and exam results. Staff can add, view, and delete records through the website.

Entity-relationship diagram

STUDENTS

PK student_id

student_name

email (unique)

enrollment_date

COURSES

PK course_id

course_code (unique)

course_name

ATTENDANCE

PK attendance_id

FK student_id

attendance_date, status

EXAM_RESULTS

PK exam_result_id

FK student_id, course_id

exam_date, grade, marks

STUDENTS 1 —— many ATTENDANCE records

STUDENTS 1 —— many EXAM_RESULTS

COURSES 1 —— many EXAM_RESULTS

Attendance refers to a student. An exam result refers to a student and a course. The website looks up names from these tables instead of storing duplicate copies.

Database tables and columns

Students

- **student_id** — INT, primary key, auto-increment.
- **student_name** — VARCHAR(100), required.
- **email** — VARCHAR(100), required and unique.
- **enrollment_date** — DATE, required.

Courses

- **course_id** — INT, primary key, auto-increment.
- **course_code** — VARCHAR(20), required and unique.
- **course_name** — VARCHAR(100), required.

Attendance

- **attendance_id** — INT, primary key, auto-increment.
- **student_id** — INT, foreign key to Students.
- **attendance_date** — DATE; one record per student per day.
- **status** — ENUM: Present or Absent.

ExamResults

- **exam_result_id** — INT, primary key, auto-increment.
- **student_id** — INT, foreign key to Students.
- **course_id** — INT, foreign key to Courses.
- **exam_date** — DATE, required.
- **grade** — VARCHAR(5), required.
- **marks** — DECIMAL(5,2), required, from 0 to 100.

DATABASE RULES AND APP

Keys, relationships & operations

Database rules

- Each table has its own auto-incrementing primary key.
- Student email and course code must be unique.
- Each student can have one attendance record per day; status is Present or Absent.
- Exam marks must be between 0 and 100.
- Attendance links to a student. Exam results link to a student and a course.
- Deleting a student removes their attendance and results. Deleting a course removes its results.

Website operations

Page	Operations
Students	Add, view, and delete student details.
Courses	Add, view, and delete courses.
Attendance	Record, view, and delete a student's attendance.
Exam results	Record, view, and delete a student's grade and marks for a course.

Technology

HTML and CSS provide the pages, PHP processes forms, and MySQL stores the data. Use `database.sql` to create the database and tables.
